2008 H2 MATHS (9740) JC2 PRELIMINARY EXAM P2- SOLUTION

Qn	Solution
1	**Inequalities**
	$\dfrac{x-6}{4x^2+x-5} \geq 1$ $\dfrac{x-6-\left(4x^2+x-5\right)}{4x^2+x-5} \geq 0$ $\dfrac{x-6-4x^2-x+5}{4x^2+x-5} \geq 0$ $\dfrac{-4x^2-1}{4x^2+x-5} \geq 0$ $\dfrac{4x^2+1}{\left(4x+5\right)\left(x-1\right)} \leq 0$ Since $4x^2+1>0$ for $x \in \mathbb{R}$, $\left(4x+5\right)\left(x-1\right)<0$ $-\dfrac{5}{4}<x<1$

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2	**AP & GP**
	For series to converge, $\left\lvert\dfrac{a^2-3a+2}{a-1}\right\rvert<1$ $\left\lvert\dfrac{(a-1)(a-2)}{a-1}\right\rvert<1$ $-1<a-2<1$ $1<a<3$ Also, $(a-1)(a-2)>0$ ($\because$ all terms in GP positive) $a<1$ or $a>2$ $\therefore$ set of values of $a=\left\{a \in \mathbb{R} : 2<a<3\right\}$ Alternative Solution: For sum to infinity to exist and all terms positive, $0<r<1$ $\Rightarrow 0<a-2<1$ $\Rightarrow 2<a<3$

	Given $a=\frac{11}{4}$. $\therefore$ $r=\frac{11}{4}-2=\frac{3}{4}$ $\frac{\frac{7}{4}\left[1-\left(\frac{3}{4}\right)^{n}\right]}{1-\frac{3}{4}}>6.999$ $1-\left(\frac{3}{4}\right)^{n}>\frac{6.999}{7}$ $\left(\frac{3}{4}\right)^{n}<1-\frac{6.999}{7}$ $n>\frac{\ln(1.4286\times10^{-4})}{\ln\frac{3}{4}}$ $n>30.776$ $\therefore$ least $n=31$

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3	**Summation**
	$\frac{1}{2x}-\frac{1}{(x-1)}+\frac{1}{2(x-2)}=\frac{(x-1)(x-2)-2x(x-2)+x(x-1)}{2x(x-1)(x-2)}$ $=\frac{x^2-3x+2-x^2+3x}{2x(x-1)(x-2)}$ $=\frac{2}{2x(x-1)(x-2)}$ $=\frac{1}{x(x-1)(x-2)}$ (verified)

$$\sum_{n=3}^{N}\frac{1}{n(n-1)(n-2)}=\frac{1}{2}\sum_{n=3}^{N}\left[\frac{1}{n}-\frac{2}{(n-1)}+\frac{1}{(n-2)}\right]$$

$$=\frac{1}{2}\left[\begin{array}{lll}\cancel{\frac{1}{3}} & -\frac{2}{2} & +\frac{1}{1}\\ +\cancel{\frac{1}{4}} & \cancel{-\frac{2}{3}} & +\frac{1}{2}\\ +\frac{1}{5} & \cancel{-\frac{2}{4}} & \cancel{+\frac{1}{3}}\\ \vdots & & \\ +\frac{1}{N-1} & \cancel{-\frac{2}{N-2}} & \cancel{+\frac{1}{N-3}}\\ +\frac{1}{N} & -\frac{2}{N-1} & \cancel{+\frac{1}{N-2}}\end{array}\right]$$

$$=\frac{1}{2}\left(\frac{1}{2}+\frac{1}{N-1}+\frac{1}{N}-\frac{2}{N-1}\right)$$

$$=\frac{1}{2}\left(\frac{1}{2}+\frac{1}{N}-\frac{1}{N-1}\right)$$

$$=\frac{1}{4}-\frac{1}{2N(N-1)}$$

Note that $\frac{1}{n^3}\le\frac{1}{n(n-1)(n-2)}=\left[\frac{1}{2n}-\frac{1}{(n-1)}+\frac{1}{2(n-2)}\right]$

$$\sum_{n=1}^{N}\frac{1}{n^3}=\frac{1}{1}+\frac{1}{2^3}+\sum_{n=3}^{N}\frac{1}{n^3}$$

$$<1+\frac{1}{8}+\frac{1}{4}-\frac{1}{2N(N-1)}.$$

$$\therefore\ \sum_{n=1}^{\infty}\frac{1}{n^3}<1+\frac{1}{8}+\frac{1}{4}=\frac{11}{8}.$$

Qn	Solution
4	**Binomial Expansions**
(a)	$(1+x)^8 = 1+8x+\frac{(8)(7)}{2!}x^2+\frac{(8)(7)(6)}{3!}x^3+\ldots$ $= 1+8x+28x^2+56x^3+\ldots$ let $x = y+ky^2$ $(1+y+ky^2)^8 = 1+8(y+ky^2)+28(y+ky^2)^2+56(y+ky^2)^3+\ldots$ From $28(y+ky^2)^2$, term in $y^3 : 28(2ky^3) = 56ky^3$ From $56(y+ky^2)^3$, term in $y^3 : 56y^3$ Given the coefficient of y^3 is zero, $56k+56=0 \Rightarrow k=-1$
(b)	$\frac{2x+3}{(x+1)(x+2)} = \frac{1}{x+1}+\frac{1}{x+2}$ $= (x+1)^{-1}+(x+2)^{-1}$ $= (1+x)^{-1}+\frac{1}{2}\left(1+\frac{x}{2}\right)^{-1}$ $= \left[1+(-1)x+\frac{(-1)(-2)}{2!}x^2+\ldots\right]+\frac{1}{2}\left[1+(-1)\left(\frac{x}{2}\right)+\frac{(-1)(-2)}{2!}\left(\frac{x}{2}\right)^2+\ldots\right]$ $= 1-x+x^2+\frac{1}{2}-\frac{x}{4}+\frac{x^2}{8}+\ldots$ $= \frac{3}{2}-\frac{5x}{4}+\frac{9x^2}{8}+\ldots$

Alternative Solution:

$$\frac{2x+3}{(x+1)(x+2)} = (2x+3)(x+1)^{-1}(x+2)^{-1}$$

$$= (2x+3)(1+x)^{-1}\left[\frac{1}{2}\left(1+\frac{x}{2}\right)^{-1}\right]$$

$$= \frac{1}{2}(2x+3)\left[1+(-1)x+\frac{(-1)(-2)}{2!}x^2+\ldots\right]\left[1+(-1)\left(\frac{x}{2}\right)+\frac{(-1)(-2)}{2!}\left(\frac{x}{2}\right)^2+\ldots\right]$$

$$= \frac{1}{2}(2x+3)(1-x+x^2+\ldots)(1-\frac{x}{2}+\frac{x^2}{4}+\ldots)$$

$$= \frac{1}{2}(2x+3)(1-\frac{3x}{2}+\frac{7x^2}{4}+\ldots)$$

$$= \frac{3}{2}-\frac{5x}{4}+\frac{9x^2}{8}+\ldots$$

Qn	Solution
5	**Differential Equations**
(a)	

$$z = x\frac{\mathrm{d}y}{\mathrm{d}x}$$

$$\frac{\mathrm{d}z}{\mathrm{d}x} = x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+\frac{\mathrm{d}y}{\mathrm{d}x} \quad \#$$

Given $x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 \Rightarrow \frac{\mathrm{d}z}{\mathrm{d}x} = 3x^2$

Hence $\int \mathrm{d}z = \int 3x^2\,\mathrm{d}x$

$$z = x^3 + c, \quad c \in \mathbb{R}$$

$$\therefore \; x\frac{\mathrm{d}y}{\mathrm{d}x} = x^3 + c$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 + \frac{c}{x}$$

$$\Rightarrow \; y = \frac{x^3}{3} + c\ln x + d, \; c, d \in \mathbb{R}$$

(b)	$\frac{d\theta}{dt} = k(1000-\theta)$ $\int \frac{1}{1000-\theta} d\theta = k\int dt$ $-\ln(1000 - \theta) = kt + C$ $1000-\theta = e^{-kt-C}$ $\theta = 1000 - Ae^{-kt}$ where $A = e^{-C}$ when $t = 0$, $\theta = 40$ $40 = 1000 - A \quad \Rightarrow A = 960$ when $t = 1$, $\theta = 160$ $160 = 1000 - 960e^{-k}$ $\Rightarrow e^{-k} = \frac{840}{960} = \frac{7}{8}$ $\therefore \theta = 1000 - 960\left(\frac{7}{8}\right)^t$

Qn	Solution
6	**Permutation and Combination**
(a) **(b)**	No of ways = $\frac{\binom{11}{3}\binom{8}{3}\binom{5}{3}\binom{2}{2}}{3!} = 15400$ No of ways = $(8-1)! \times (4)(3) \times 2 = 120960$

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7	**Sampling Method**
(i)	Simple random sampling is • free from bias • relatively small sample size, hence easy to conduct • analysis of data relatively easy Disadvantage: May not be easy to get access to members who have been chosen for the sample
(ii)	Systematic sampling • Simple to use for large number of walk-in customers • Sample obtained more evenly spread over the spectrum of walk-in customers • Relatively easy to conduct The hobby shop could decide on an interval to select a walk-in customer for the survey, for example, every 20th customer. It then chooses a random start between the numbers 1 and 20. This will determine the first customer to be surveyed and every 20th customer thereafter will be surveyed.

Qn	Solution
8	**Binomial Distribution**
(i)	Let X be the number of students out of n who use AIKON brand mobile phones. When $n = 30$, $$X \sim B\left(30, \frac{1}{5}\right)$$ Expected number of students who use AIKON brand phone $= (30)\left(\frac{1}{5}\right) = 6$ $\mathrm{P}(X = 5) = 0.172$
(ii)	$\mathrm{P}(X \geq 1) > 0.99$ $\Rightarrow 1 - \mathrm{P}(X = 0) > 0.99$ $\Rightarrow \mathrm{P}(X = 0) < 0.01$ $\Rightarrow \left(\frac{4}{5}\right)^n < 0.01$ $\Rightarrow n > \dfrac{\lg(0.01)}{\lg\left(\frac{4}{5}\right)} = 20.6$ Hence, least $n = 21$.

Qn	Solution
9	**Estimation and Hypothesis Testing**
(i)	Unbiased estimate for population mean = $\bar{x} = \frac{\sum x}{20} = 1.96$ Unbiased estimate for population variance = $s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{\left(\sum x\right)^2}{n}\right) = \frac{1}{19}\left(77.022 - \frac{(39.2)^2}{20}\right)$ = 0.01
(ii)	$H_0 : \mu = 2$ $H_1 : \mu < 2$ Assumption: $X \sim N\left(\mu, \sigma^2\right)$ Test statistic: $T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ Level of significance: $\alpha\%$ Critical region: Reject H_0 if *p*-value < $\frac{\alpha}{100}\%$ Assuming H_0 is true, from the GC, *p*-value = 0.044797 = 0.0488 (3 s.f) Set of values of $\alpha = \left\{\alpha \in \mathbb{R} : 4.48 \le \alpha \le 100\right\}$

Qn	Solution
10	**Probability**
(i)	$A \cap B$ represents the event the card taken is red and numbered 1. P($A \cap B$) = $\frac{1}{40}$. Note: If students use the formula that $\mathrm{P}\left(A \cap B\right) = \mathrm{P}\left(A\right) \times \mathrm{P}\left(B\right)$, they need to prove that event *A* and event *B* are independent.

(ii)	$A \cup B$ represents the event the card taken is either red or numbered 1. $\text{P}(A \cup B) = \text{P}(A) + \text{P}(B) - \text{P}(A \cap B)$ $= \frac{10}{40} + \frac{4}{40} - \frac{1}{40}$ $= \frac{13}{40}$
(iii)	$\text{P}(A' \mid B') = \frac{P\left(A' \cap B'\right)}{P(B')}$ $= \frac{1 - \frac{13}{40}}{\frac{36}{40}}$ $= \frac{3}{4}$
	Required probability $= \binom{4}{3}\left(\frac{1}{40}\right)\left(\frac{1}{39}\right)\left(\frac{1}{38}\right)3! \times 10$ $= 0.00405$

Qn	Solution
11	**Poission**
(i) **(ii)**	Let X be the no of requests for the hire of a truck on a day. $X \sim \text{Po}(2)$ Prob at most 1 truck not hired out $= \text{P}(X \geq 2)$ $= 1 - \text{P}(X \leq 1)$ $= 0.59399$ $= 0.594(3\text{sf})$ Prob that requests can be met on a given day $= P(X \leq 3)$ $= 0.85712$ $= 0.857(3\text{sf})$
	Prob that a particular truck is not used $= \text{P}(X = 0) + \frac{2}{3}\text{P}(X = 1) + \frac{2}{3} \times \frac{1}{2}\text{P}(X = 2)$ $= 0.40601$ $= 0.406\ (3\text{sf})$

	Let the no of trucks be n $\mathrm{P}(X > n) < 0.01$ $\mathrm{P}(X \le n) > 0.99$ From GC, $\mathrm{P}(X \le 5) = 0.98344 < 0.99$ $\mathrm{P}(X \le 6) = 0.99547 > 0.99$ least no of trucks is 6

Qn	Solution
12	**Normal Distribution**
	Let F = the length, in cm, of a floor tile. $F \sim \mathrm{N}\left(30, 0.2^2\right)$ Let L = the length, in cm, of a wall tile. $L \sim \mathrm{N}\left(10, 0.15^2\right)$
(i)	$F_1 + F_2 + F_3 + F_4 + F_5 \sim \mathrm{N}(150, 0.2)$ $\mathrm{P}(150.5 < F_1 + F_2 + F_3 + F_4 + F_5 < 151.5)$ $= 0.13138$ $= 0.131$ (3 s.f.)
(ii)	$S = L_1 + L_2 + L_3 + L_4 + L_5 + L_6 \sim \mathrm{N}(60, 0.135)$ $W = 2F \sim \mathrm{N}\left(60, 0.4^2\right)$ $\mathrm{P}(S < W + 1)$ $= \mathrm{P}(S - W < 1) \qquad S - W \sim \mathrm{N}(0, 0.295)$ $= 0.96720$ $= 0.967$
(iii)	$\overline{F} = \dfrac{F_1 + F_2 + \ldots + F_{25}}{25} \sim \mathrm{N}\left(30, 0.04^2\right)$ $\mathrm{P}\left(\left\lvert\overline{F} - 30\right\rvert \ge a\right) = 0.02$ $\Rightarrow \mathrm{P}\left(\lvert Z\rvert \ge \dfrac{a}{0.04}\right) = 0.02$ $\Rightarrow \mathrm{P}\left(Z \le \dfrac{-a}{0.04}\right) + \mathrm{P}\left(Z \ge \dfrac{a}{0.04}\right) = 0.02$ $\Rightarrow \mathrm{P}\left(Z \le \dfrac{-a}{0.04}\right) = 0.01$ $\Rightarrow \dfrac{-a}{0.04} = -2.32635 .\ \therefore a = 0.0931$ (3 s.f.)

Qn	Solution
13	**Regression and Correlation**
(i)	Suitable to use h on s since s is clearly, the controlled [independent] variable.
(ii)	Let $x = s - 20$ and $y = h - 100$ Regression line of y on x is: $y - \bar{y} = b\left(x - \bar{x}\right)$ where $\bar{x} = \frac{1}{n}\sum x = \frac{143}{15} = 9.5333$, $\bar{y} = \frac{1}{n}\sum y = \frac{391}{15}$=26.067 and $b = \dfrac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{\left(\sum x\right)^2}{n}} = \dfrac{484 - \frac{143 \times 391}{15}}{2413 - \frac{(143)^2}{15}} = -3.0899$ $\therefore y - 26.067 = -3.0899\left(x - 9.5333\right) \Rightarrow y = 55.523 - 3.0899x$ Thus the line of h on s is: $h - 100 = 55.523 - 3.0899\left(s - 20\right) \Rightarrow h = 217.32 - 3.0899s \Rightarrow h = 217 - 3.09s$ [3 sf]
(iii) **(iv)** **(v)**	For every extra revolution/minute the life of the drill reduces by about 3 hours. Find the product moment correlation coefficient, r and if $r \approx -1$ then the calculated regression line may be a good fit to the data. OR If the raw data is available then a scatter diagram could be drawn with the regression line on it. If the points lie close to the regression line, then the line is said to fit the data well. At $h = 125, -3.0899s + 217.32 = 125 \Rightarrow s = \frac{217.32 - 125}{3.0899}$ =29.9 $\approx$ 30 revs/minute